NATIONAL FORENSIC SCIENCES UNIVERSITY, DELHI CAMPUS

P3: Vulnerability assessment and penetration testing

PRACTICAL 2: Reconnaissance &Information Gathering

1. Aim

To perform reconnaissance and information gathering on a domain using OSINT tools, DNS utilities, port-scanning tools, and web-probing tools to identify subdomains, resolve active hosts, and discover exposed network services.

2. Tools required

- Kali Linux
- Sub finder—passive subdomain enumeration
- dig—DNS record lookup
- DNSx — DNS resolution
- Naabu—port scanning
- HTTPX — HTTP/HTTPS service probing
- grep and wc—filtering and counting results

3. Theory

Reconnaissance is the initial phase of penetration testing in which information about a target's publicly exposed infrastructure is collected. Passive reconnaissance uses publicly available sources, while active reconnaissance sends queries or connections to target systems.

Subdomain enumeration identifies hostnames associated with a domain. DNS validation checks whether those hostnames resolve to IP addresses. Port scanning

identifies accessible network services, and web probing checks which hosts serve HTTP or HTTPS content.

4. Procedure: Step-by-step practical

Phase 1 — Passive subdomain enumeration

Step 1: Open Kali Linux Terminal

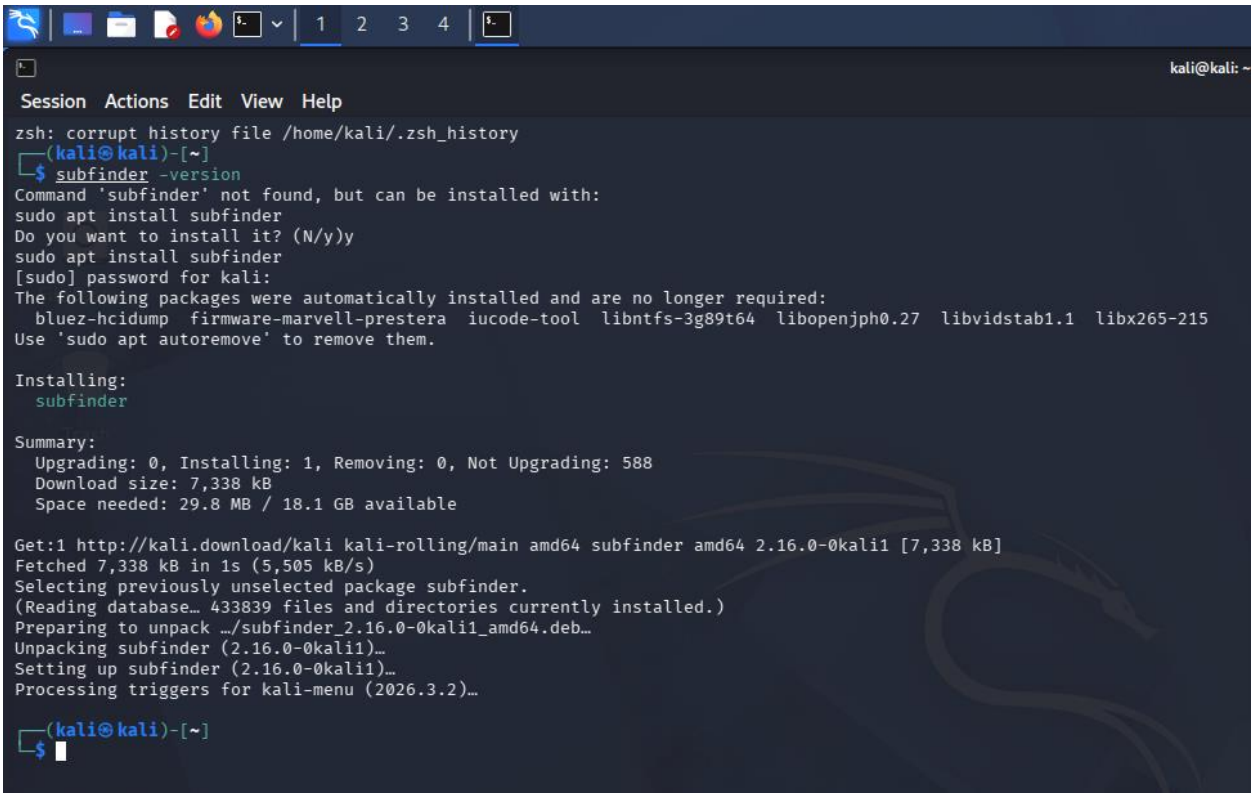
Open your Kali Linux virtual machine and launch the terminal using `Ctrl + Alt + T`.

Step 2: Check whether Subfinder is installed

Run:

```
subfinder -version
```

If it is not installed, install it using the official Project Discovery installation instructions.



Step 3: Enumerate subdomains

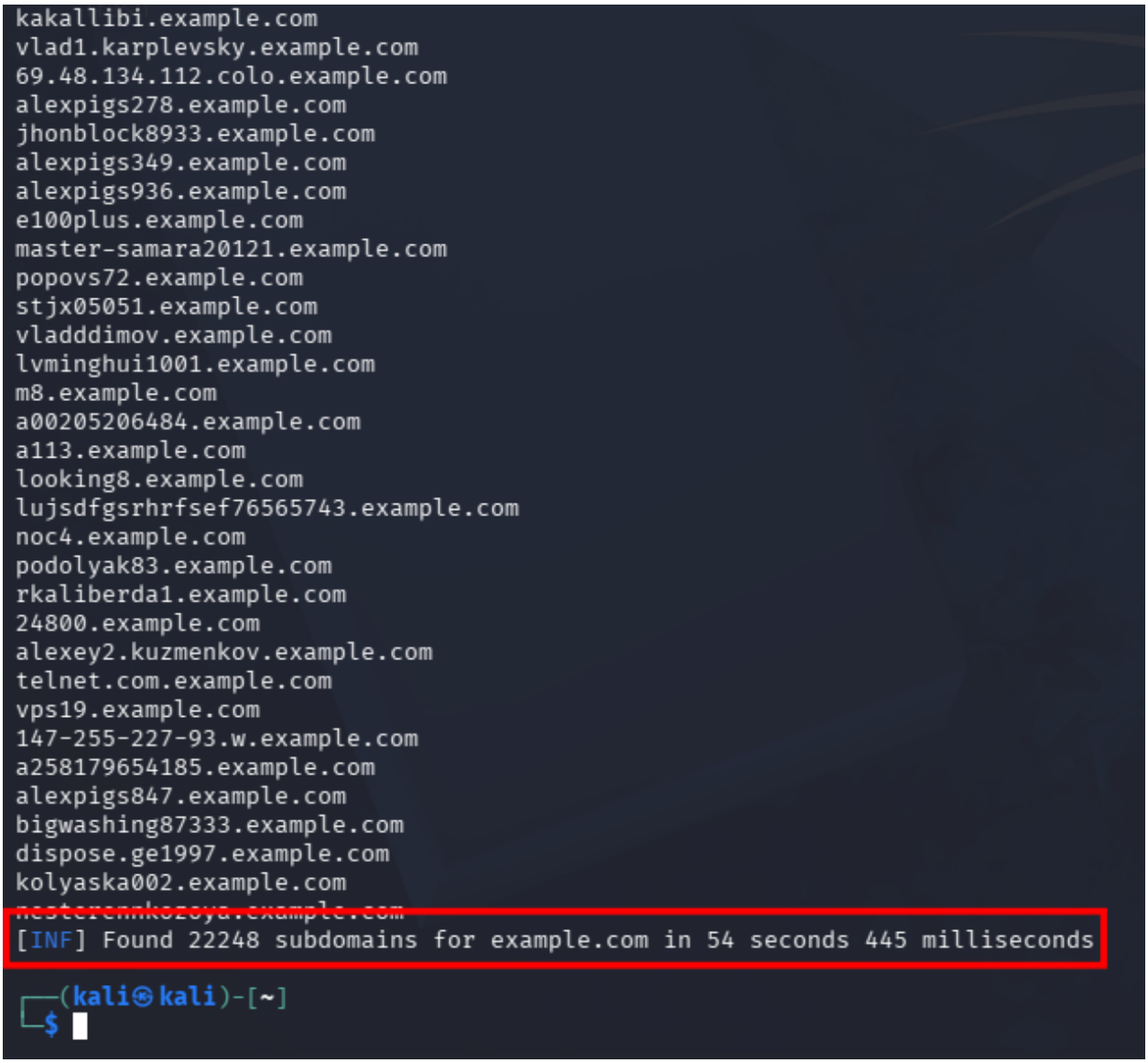
Replace the example target with a domain you are authorized to assess.

```
subfinder -d example.com -v -o passive_subdomains.txt
```

Explanation:

- `-d` specifies the target domain.
- `-v` enables verbose output.
- `-o` saves the discovered subdomains to a file.
- `passive_subdomains.txt` is the output filename.

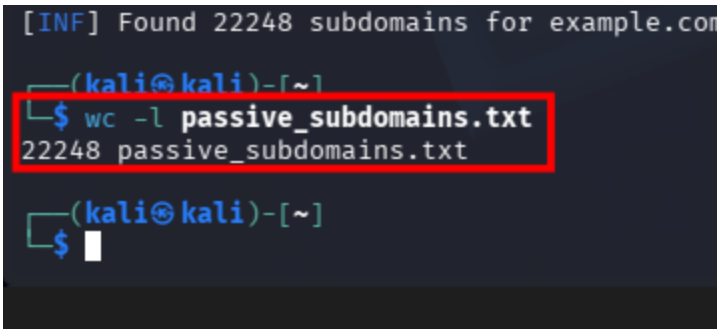




Step 4: Count the discovered subdomains

```
wc -l passive_subdomains.txt
```

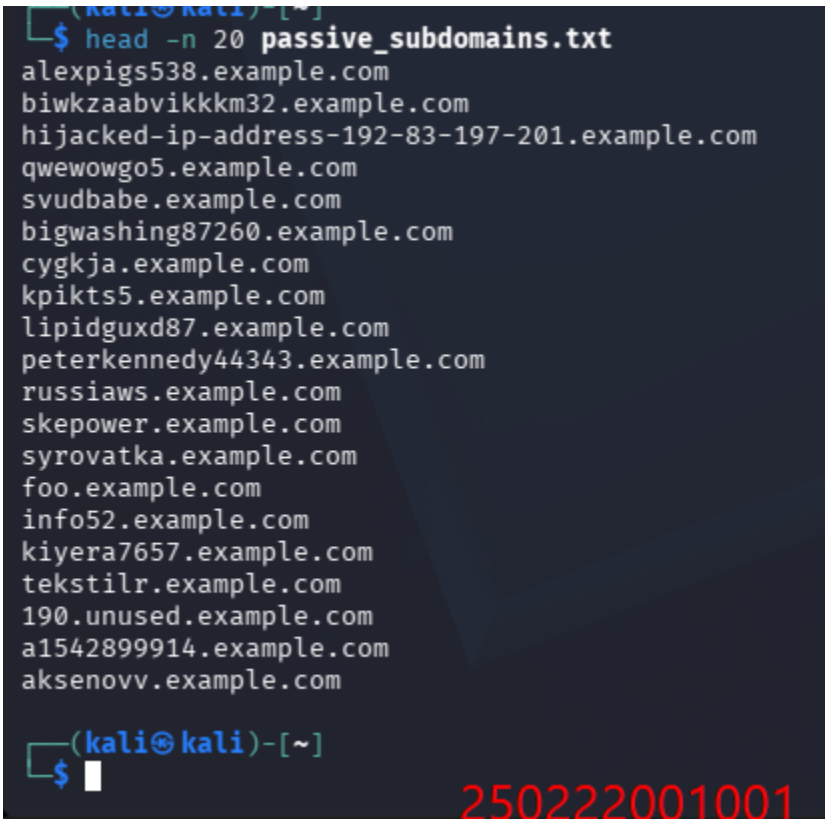
This displays the number of lines in the output file.



Step 5: Display the first 20 results

```
head -n 20 passive_subdomains.txt
```

Expected output: a list of discovered hostnames, if any are returned by the available passive data sources.



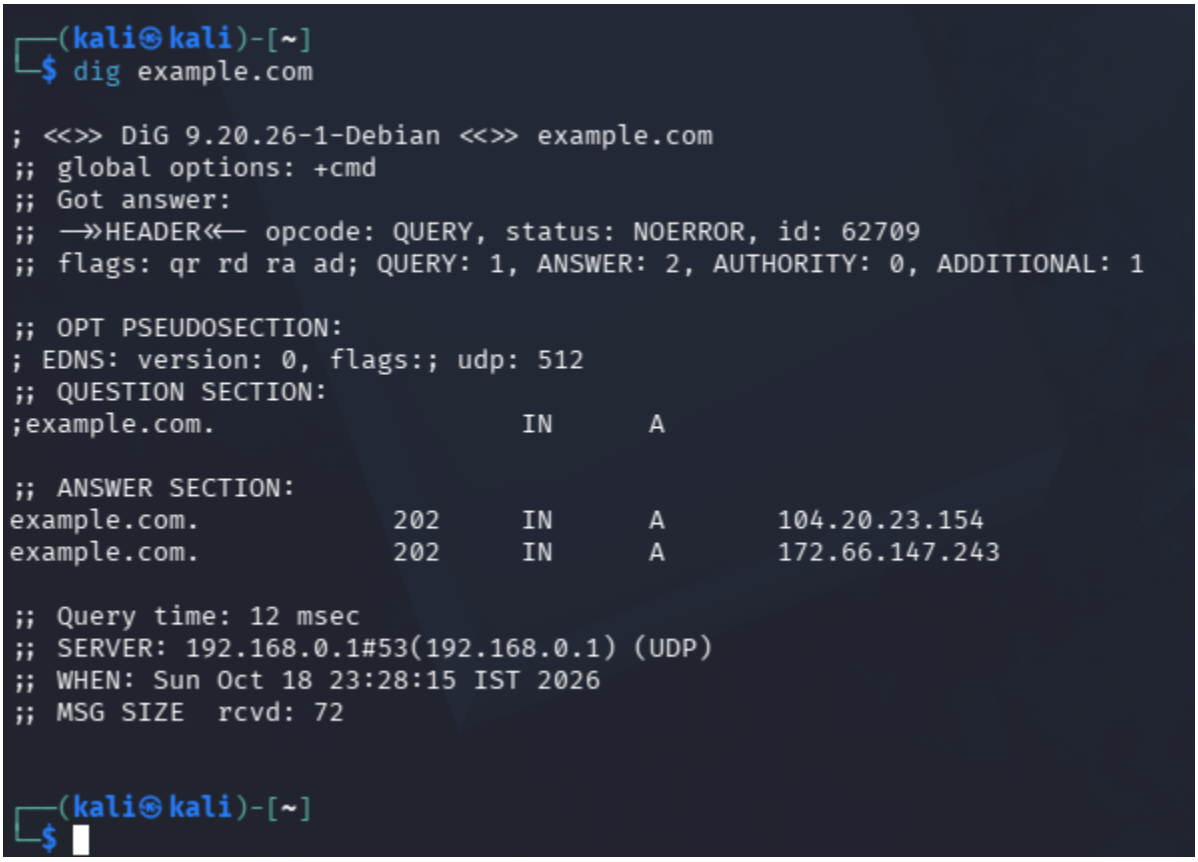
Phase 2 — DNS resolution and validation

The objective is to check which discovered hostnames currently resolve through DNS.

Step 6: Check DNS using `dig`

Run:

`dig example.com`



Then test a known hostname:

`dig` www.example.com

```
┌──(kali㉿kali)-[~]
└─$ dig www.example.com

; <<>> DiG 9.20.26-1-Debian <<>> www.example.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 51815
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 2, AUTHORITY: 0, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;www.example.com.               IN      A

;; ANSWER SECTION:
www.example.com.        295     IN      A       104.20.23.154
www.example.com.        295     IN      A       172.66.147.243

;; Query time: 12 msec
;; SERVER: 192.168.0.1#53(192.168.0.1) (UDP)
;; WHEN: Sun Oct 18 23:29:45 IST 2026
;; MSG SIZE  rcvd: 76


┌──(kali㉿kali)-[~]
└─$
```

To test a deliberately unlikely hostname:

`dig nonexistent-unique-12345.example.com`

```
└─$ dig nonexistent-unique-12345.example.com

; <<>> DiG 9.20.26-1-Debian <<>> nonexistent-unique-12345.example.com
;; global options: +cmd
;; Got answer:
;; ->>HEADER<<- opcode: QUERY, status: NOERROR, id: 61892
;; flags: qr rd ra ad; QUERY: 1, ANSWER: 0, AUTHORITY: 1, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 512
;; QUESTION SECTION:
;nonexistent-unique-12345.example.com. IN A

;; AUTHORITY SECTION:
example.com.            1800    IN      SOA     elliott.ns.cloudflare.com. dns.cloudflare.com. 2416374680 10000 2400 604800 1800

;; Query time: 104 msec
;; SERVER: 192.168.0.1#53(192.168.0.1) (UDP)
;; WHEN: Sun Oct 18 23:31:18 IST 2026
;; MSG SIZE  rcvd: 127
```

Interpret the output:

- NOERROR means the DNS query completed successfully; check the answer section to determine whether a record exists.
- NXDOMAIN means the queried name does not exist in DNS.

- An A record typically maps a hostname to an IPv4 address.

Step 7: Check whether DNSx is installed

Step 8: Resolve the enumerated hostnames

For a domain you are authorized to assess:

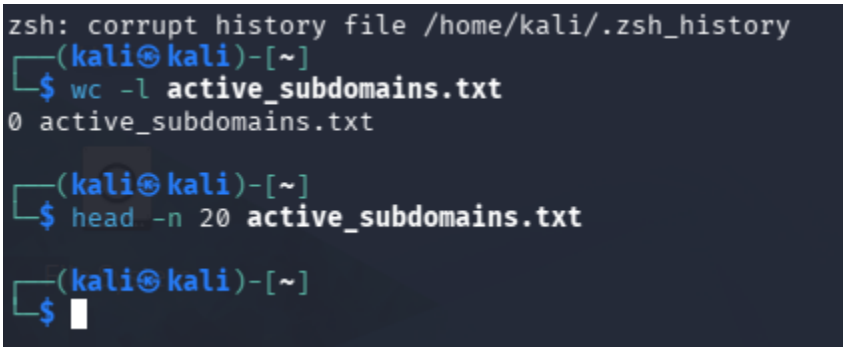
```
cat passive_subdomains.txt | dnsx -a -resp -o active_subdomains.txt
```

Explanation:

- `cat` reads the input file.
- `|` passes the hostnames to DNSx.
- `-a` requests IPv4 address records.
- `-resp` includes DNS response information.
- `-o` saves the results.



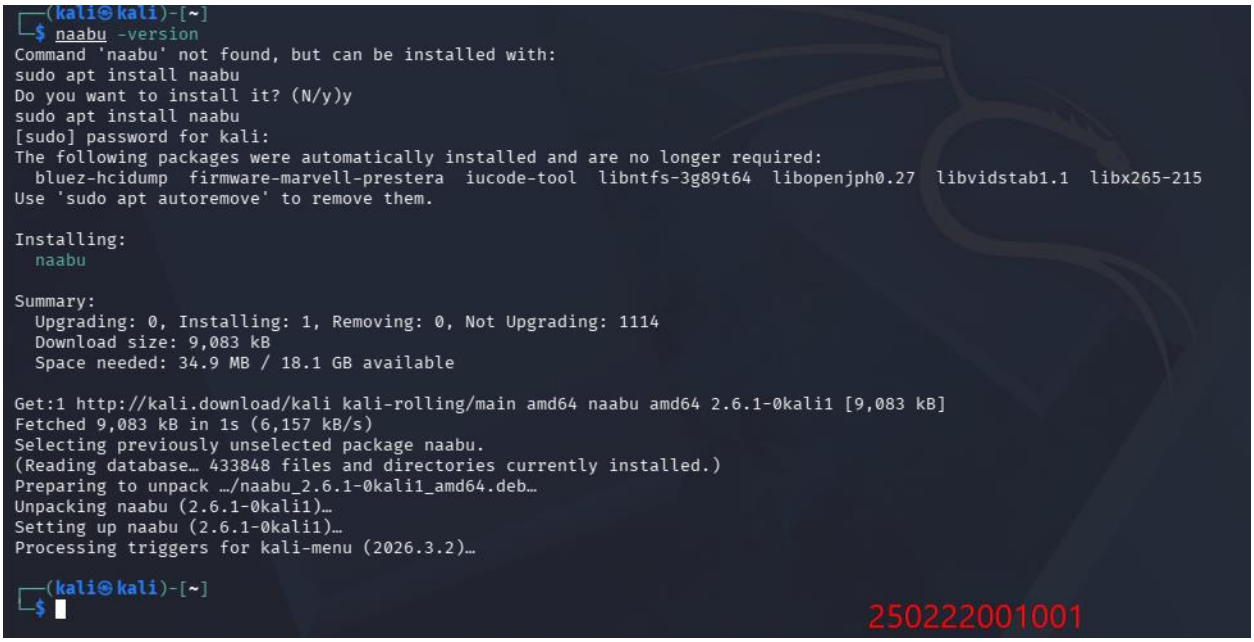
Step 9: Verify the output count



Phase 3 — Port scanning and web probing

This phase actively connects to systems, so perform it only on your own lab machine or an explicitly authorized target.

Step 10: Check Naabu installation



Step 11: Test port scanning on your own Kali machine

```
naabu -host 127.0.0.1 -p 80,443,8080,8443
```

Explanation:

- `-host` specifies the host being tested.
- `127.0.0.1` is the loopback address of your own machine.
- `-p` specifies the ports to check.

If a local service is listening, Naabu may report an open port. If none of the listed ports is open, it may return no open-port results. That is a valid observation.



5. Observation

Use these observation points in your lab record and fill in the actual results obtained on your system.

1. Subfinder was used to collect subdomains from passive sources.
2. The output was stored in `passive_subdomains.txt`.
3. DNS queries were performed using `dig`, and the results were examined for valid records and nonexistent hostnames.
4. DNSx was used to identify hostnames with resolvable IPv4 records.
5. Naabu was used to check selected ports on a local test host.
6. HTTPX was used to inspect an HTTP endpoint and obtain response metadata.

6. Result

Reconnaissance and information gathering were performed using Subfinder, `dig`, DNSx, Naabu . The collected data was saved in output files, and DNS resolution, port availability and HTTP response information were examined. The actual number of discovered subdomains and open ports depends on the environment and test target.

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